

Military Technical Academy (MTA)



**EDR-REDNet: An Edge-Guided RED-CNN
Extension for Structure-Preserving Low-Dose
CT Denoising**

Information technology

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Hanoi, 06/2026

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Abstract

Low-dose computed tomography (LDCT) reduces radiation exposure but introduces quantum noise and artifacts that may obscure fine anatomical boundaries. Conventional denoising networks often optimize global image fidelity metrics such as PSNR and SSIM, which may favor overly smooth outputs and fail to reflect local edge preservation. This study proposes EDR-REDNet, a compact edge-aware extension of RED-CNN for structure-preserving LDCT image denoising. The model integrates fixed four-direction Sobel edge guidance, edge-dilated residual bottleneck processing, and a Sobel-gradient supervision term combined with a robust Charbonnier reconstruction loss. A controlled ablation study was conducted against RED-CNN, an EdgeBlock variant, and a Sobel-input variant. The primary endpoint was patient-level Edge.SSIM, while PSNR, global SSIM, parameter count, MACs, and inference time were treated as secondary endpoints. On nine held-out patients, EDR-REDNet achieved the highest Edge.SSIM, improving from 0.7608 ± 0.1308 for RED-CNN to 0.7879 ± 0.1291 , with consistent improvement across all patients and statistically significant Wilcoxon signed-rank results. RED-CNN retained slightly higher PSNR and global SSIM, indicating a structure-preserving trade-off rather than universal superiority. EDR-REDNet increased inference time by approximately $1.23\times$ compared with RED-CNN. These findings suggest that explicit edge guidance and gradient supervision can reduce over-smoothing in LDCT denoising, but larger multi-center validation, radiologist reader studies, radiomic stability analysis, and downstream diagnostic evaluation are required before clinical use.

1 Introduction

1.1 Problem Statement

Computed tomography (CT) is an indispensable imaging modality in modern medicine because it provides high-resolution cross-sectional anatomical information for diagnosis, staging, treatment planning, interventional guidance, and longitudinal follow-up. However, CT imaging relies on ionizing radiation, and repeated examinations or large-scale screening workflows raise concerns regarding cumulative radiation exposure. Under the principle of keeping radiation exposure as low as reasonably achievable, low-dose computed tomography (LDCT) has become an important strategy for reducing patient dose while preserving the clinical utility of CT imaging [1–3].

The clinical importance of LDCT is particularly evident in lung cancer screening. Large randomized trials have shown that LDCT screening can reduce lung cancer mortality in high-risk populations, making LDCT a central imaging tool for early detection programs [4]. Nevertheless, lowering radiation dose typically reduces the number of detected photons and degrades the statistical quality of the acquired data. As a result, reconstructed LDCT images often contain increased quantum noise, streak artifacts, reduced contrast, unstable Hounsfield-unit values, and blurred anatomical boundaries. In lung imaging, these degradations are clinically relevant because small pulmonary nodules, vessel margins, bronchi, fissures, and subtle parenchymal structures may occupy only a limited spatial extent and may be difficult to distinguish from noise or texture loss.

A wide range of LDCT restoration methods has been developed to address this problem. Projection-domain and iterative reconstruction methods can incorporate acquisition physics and noise statistics, but they often require access to raw projection data, vendor-specific reconstruction pipelines, or substantial computational resources. In contrast, image-domain denoising methods operate directly on reconstructed CT images and are therefore easier to integrate into existing clinical and research workflows. Deep learning-based image-domain methods, including convolutional neural networks, encoder-decoder architectures, generative adversarial networks, and transformer-based models, have shown strong performance in LDCT denoising and reconstruction tasks [1,2,5]. Among these methods, RED-CNN remains an important reference architecture because it is compact, reproducible, and has remained competitive in benchmark evaluations.

Despite substantial progress, LDCT denoising remains challenging because visual noise suppression and diagnostic structure preservation are not equivalent objectives. Conventional full-reference image-quality metrics such as peak signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) are widely reported, but they do not fully characterize clinically relevant boundary preservation. PSNR is primarily driven by pixel-wise fidelity and may favor overly smooth outputs, whereas global SSIM summarizes broad structural similarity and may not isolate small edge-level differences. Recent benchmarking work has further emphasized that many LDCT denoising algorithms show statistically similar performance under standard metrics and that reported improvements can be difficult to verify without rigorous, reproducible evaluation protocols [2]. These observations motivate a more specific question: can an LDCT denoising model improve edge-structure preservation even when global PSNR or SSIM is not maximized?

Edge preservation is particularly important in medical CT because

anatomical boundaries often correspond to clinically meaningful structures. Loss of edge information may reduce the visibility of small nodules, vessel contours, lesion margins, and organ boundaries. Prior work has explored edge-aware and perceptual constraints for LDCT denoising. For example, Gholizadeh-Ansari et al. proposed a deep network incorporating dilated convolutions, residual learning, a non-trainable edge detection layer, and perceptual loss, showing that edge-aware components can help preserve structural details while limiting over-smoothing [6]. However, further work is needed to evaluate edge-guided modifications within a reproducible RED-CNN-based benchmark setting, with controlled ablations, patient-level statistics, and explicit reporting of computational overhead.

In this study, we propose EDR-REDNet, an Edge-Dilated Residual RED-CNN for structure-preserving LDCT image denoising. EDR-REDNet extends a RED-CNN-style encoder-decoder with three edge-aware components: a fixed four-direction Sobel edge-guidance layer, edge-dilated residual bottleneck blocks, and a Sobel-gradient supervision term combined with a robust reconstruction loss. The fixed Sobel layer provides transparent, non-trainable edge cues; the edge-dilated residual blocks enlarge contextual perception at the bottleneck without excessive parameter growth; and the Sobel-gradient loss directly aligns the training objective with boundary preservation. The proposed model is intentionally designed as an image-domain method, preserving the practical deployment advantages of RED-CNN while introducing explicit edge-structure guidance.

The purpose of this work is not to claim universal superiority over RED-CNN or to maximize PSNR and global SSIM. Instead, the study evaluates whether targeted edge-aware architectural guidance and gradient-based supervision can reduce over-smoothing and improve edge-structure preservation. To test this hypothesis, we conduct a controlled ablation study comparing four variants: a pretrained RED-CNN baseline, a version with edge-dilated residual bottleneck blocks, a version with Sobel input guidance, and the full EDR-REDNet model. The primary endpoint is patient-level Edge_SSIM, while PSNR, global SSIM, parameter count, multiply – accumulate operations, and inference time are treated as secondary metrics.

The main contributions of this work are as follows:

We introduce EDR-REDNet, a compact edge-aware extension of RED-CNN that integrates fixed Sobel guidance, edge-dilated residual bottleneck processing, and Sobel-gradient supervision for LDCT image denoising. We formulate LDCT denoising as a structure-preserving restoration problem rather than only a pixel-wise noise suppression problem. We perform a controlled ablation study to isolate the effects of edge-dilated residual blocks, So-

bel input guidance, and Sobel-gradient loss. We evaluate performance at the patient level using Edge_SSIM as the primary endpoint, avoiding the overly optimistic interpretation that can arise from treating correlated CT slices as independent samples. We report statistical significance and efficiency metrics to support reproducible interpretation. In the current evaluation, the full model improves Edge_SSIM in all nine held-out patients and reaches statistical significance against the RED-CNN baseline and partial variants, while introducing moderate computational overhead.

Existing LDCT denoising studies often report PSNR and SSIM, but these metrics may not adequately reflect small boundary preservation in lung CT. Moreover, improvements are frequently reported at slice level without patient-level statistical analysis. This motivates an edge-aware, RED-CNN-based model evaluated using patient-level Edge SSIM and controlled ablations.

The remainder of this paper is organized as follows. Section 2 reviews related LDCT denoising methods, RED-CNN-based restoration, and edge-aware learning. Section 3 describes the proposed EDR-REDNet architecture, loss function, ablation variants, and evaluation protocol. Section 4 presents quantitative results, patient-level Edge_SSIM analysis, statistical testing, and computational efficiency. Section 5 discusses the structure-preserving trade-off, limitations, and future validation requirements. Section 6 concludes the study.

2 Related Work

2.1 Background

2.1.1 Low-dose CT denoising and reconstruction methods

Low-dose CT image restoration has been studied through several methodological families, which can broadly be grouped according to the domain in which the correction is performed: projection-domain processing, iterative reconstruction, image-domain post-processing, and more recent dual-domain learning approaches. Projection-domain methods operate on raw sinogram data before image reconstruction. Since photon statistics are more directly modeled in the projection domain, these methods can exploit the physical formation of LDCT noise and artifacts. However, access to raw projection data is usually restricted by scanner vendors, and the implementation of projection-domain correction often requires integration with proprietary reconstruction pipelines.

Iterative reconstruction methods incorporate physical models, scanner geometry, photon statistics, and prior assumptions into an optimization framework. These approaches can provide strong noise suppression and artifact reduction, but they are computationally expensive and often scanner dependent. Their clinical deployment may therefore be limited by reconstruction time, vendor-specific implementation, and the difficulty of reproducing the same algorithm across institutions. In contrast, image-domain methods operate directly on reconstructed CT images. They do not require raw sinogram access and can be implemented as post-processing modules after standard scanner reconstruction. This makes image-domain LDCT denoising attractive for research and practical deployment, especially when projection data are unavailable.

Deep learning has substantially advanced image-domain LDCT restoration. Recent reviews classify LDCT denoising networks into CNN-based, encoder-decoder-based, GAN-based, and Transformer-based methods. CNN-based models efficiently extract local image features and remain competitive for denoising tasks. Encoder-decoder architectures learn hierarchical representations and are commonly optimized with pixel-wise losses such as MSE or L1. GAN-based methods introduce adversarial or perceptual objectives to improve visual realism, whereas Transformer-based methods are designed to capture long-range dependencies and global contextual information.

Dual-domain and sinogram-informed methods have also been proposed to bridge the gap between physics-based reconstruction and image-domain learning. For example, Sinogram Inner-Structure Transformer exploits the

internal structure of sinogram data and uses self-attention to model relationships between projection views. Such methods are attractive because they explicitly incorporate acquisition-domain information; however, they still depend on the availability of projection data or a differentiable reconstruction pathway. For many practical LDCT datasets, including benchmark settings and hospital archives where only reconstructed images are available, image-domain methods remain the most accessible option.

In this context, the present study focuses on image-domain LDCT denoising. This choice is not intended to replace projection-domain or dual-domain reconstruction, but to develop a deployable and reproducible edge-preserving extension within an existing RED-CNN-style denoising framework.

2.1.2 RED-CNN-based restoration

RED-CNN is one of the most widely used image-domain baselines for LDCT denoising. It follows a residual encoder–decoder design in which convolutional encoder layers compress the LDCT input into latent features, and decoder layers reconstruct the denoised image. Skip connections are used to transfer image information from encoder layers to corresponding decoder layers, helping the model preserve low-level structures during reconstruction. Reviews of LDCT denoising commonly present RED-CNN as a representative encoder–decoder architecture and distinguish its skip-connection strategy from U-Net-style concatenation, noting that RED-CNN variants often use summation-based feature transfer.

RED-CNN remains important not only because of its historical role, but also because it is compact, reproducible, and difficult to surpass consistently under conventional global metrics. The 2024 LDCT benchmark study by Eulig et al. emphasized that many published deep learning denoising methods are difficult to compare because of inconsistent datasets, hyperparameter choices, and evaluation protocols. Their standardized evaluation further showed that many newer methods achieve statistically similar performance to older baselines, highlighting the need for rigorous benchmarking rather than uncontrolled claims of improvement.

This benchmark context motivates the use of RED-CNN as the primary reference model in this study. A new method should not be evaluated only by reporting a higher average metric on a non-standard split; instead, it should be tested under a controlled ablation design with consistent preprocessing, patient-level aggregation, statistical testing, and efficiency reporting. In the proposed EDR-REDNet study, RED-CNN is therefore treated as a strong baseline, not as a weak comparator. This is important because the goal of

EDR-REDNet is not to claim universal superiority over RED-CNN, but to test whether a targeted edge-aware modification can improve edge-structure preservation.

Several later LDCT denoising models also build upon RED-CNN-like components. For example, review evidence notes that DU-GAN uses a RED-CNN sub-network in its generator and combines image-domain and gradient-domain losses to balance noise suppression, artifact reduction, and structural-detail preservation. This suggests that RED-CNN remains a useful backbone for developing structure-preserving variants, especially when the goal is to introduce targeted architectural or loss-function modifications without moving to much larger models

2.1.3 Edge-aware learning for structure preservation

Although LDCT denoising methods often report PSNR, SSIM, and RMSE, these global metrics do not fully describe whether clinically relevant local structures are preserved. PSNR is strongly tied to mean squared error and may reward smooth outputs, while SSIM summarizes broad structural similarity and may not isolate small vessel boundaries, fissures, nodule margins, or local edge patterns. Recent reviews explicitly note that LDCT denoising should not only suppress noise, but should also preserve image edges, maintain global contrast, avoid texture loss, and avoid introducing new artifacts.

This issue is especially important in chest CT. Edges in lung CT may correspond to vessel walls, bronchial boundaries, fissures, nodule margins, pleural interfaces, and lesion boundaries. Excessive smoothing can reduce the visibility of these structures, whereas excessive sharpening may preserve noise edges or generate artificial details. Therefore, structure-preserving LDCT denoising requires a balance between noise suppression and conservative edge preservation.

Edge-aware deep learning methods attempt to address this problem by explicitly guiding the model with gradient or boundary information. Gholizadeh-Ansari et al. proposed an LDCT denoising network using dilated convolutions, residual learning, a non-trainable edge detection layer, and a combination of MSE and perceptual loss. Their edge detection layer extracts edges in horizontal, vertical, and diagonal directions, while dilated convolutions expand contextual perception without substantially increasing the number of layers. The study argued that these components can help preserve structural details and reduce over-smoothing with only minimal changes in model complexity.

Other methods incorporate gradient-domain or perceptual supervision.

For instance, DU-GAN combines image-domain MSE with gradient-domain MSE and adversarial learning, which has been reported to moderate over-smoothing and preserve structural details. Transformer-based methods can capture long-range context, but recent review analysis also notes that some Transformer variants may lose local texture detail unless complemented by CNN-based local enhancement modules. This supports the idea that global context alone is insufficient for structure-preserving LDCT denoising; local edge and texture constraints remain necessary.

Despite the promise of perceptual losses, their use in CT denoising should be considered carefully. Perceptual losses based on natural-image networks such as VGG were not designed for CT intensity physics or Hounsfield-unit fidelity. They may improve visual appearance, but their feature representations are not inherently tied to grayscale CT boundary structure. For this reason, the present work uses Sobel-gradient supervision instead of natural-image perceptual supervision. Sobel operators are transparent, deterministic, lightweight, and directly related to local gradient structure in CT images.

2.1.4 Benchmarking, metrics, and reproducibility

A key challenge in LDCT denoising research is that reported performance gains are often sensitive to dataset split, preprocessing, hyperparameter tuning, and metric choice. Eulig et al. identified several flaws in common evaluation protocols, including lack of standardized benchmark datasets, inadequate metrics with limited relation to diagnostic value, unfair hyperparameter choices for reference methods, and inconsistent implementation details. Their results showed that many methods perform similarly under fair comparison, and that progress over earlier baselines is often smaller than individual papers suggest.

These observations have two implications for the present work. First, RED-CNN should be treated as a serious baseline because it remains competitive under standardized evaluation. Second, a proposed edge-aware method should be evaluated according to the endpoint it is designed to improve. If a model is designed to preserve local anatomical boundaries, then PSNR and global SSIM alone are insufficient. They remain useful secondary metrics, but they should not be the sole basis for assessing structure preservation.

The EDR-REDNet evaluation therefore follows a benchmark-oriented strategy. It compares a RED-CNN baseline with controlled ablation variants under the same preprocessing and evaluation workflow. The primary endpoint is patient-level Edge_SSIM, while PSNR, global SSIM, parameter count, MACs, and inference time are reported as secondary measures.

Patient-level aggregation is used because slices from the same CT volume are correlated and should not be treated as fully independent samples. This design is intended to reduce inflated statistical confidence and to focus the analysis on whether the edge-aware components produce consistent patient-level gains.

2.1.5 Position of the proposed EDR-REDNet

3 Materials and Methods

3.1 Study design

This study was designed as a retrospective computational evaluation of an edge-aware LDCT denoising model. The central hypothesis was that explicit edge guidance and Sobel-gradient supervision could improve edge-structure preservation compared with a RED-CNN baseline and partial ablation variants. The primary endpoint was patient-level Edge_SSIM. PSNR, global SSIM, model size, multiply-accumulate operations (MACs), and inference time were treated as secondary endpoints. The study was conducted within a local ldct-benchmark-based research workflow to ensure consistent preprocessing, model loading, evaluation, and statistical testing across all compared variants.

The proposed model, named EDR-REDNet, was evaluated against three comparator variants: a pretrained RED-CNN baseline, a model with edge-dilated residual bottleneck blocks, and a model with Sobel edge guidance but without the complete edge-supervised objective. The purpose of this design was to isolate the contribution of each proposed component and to determine whether the full combination of architecture and loss function was required to improve edge preservation.

3.2 Dataset and preprocessing

The final evaluation used nine held-out patients available in the local benchmark result set: L150, L110, L232, L134, C002, C004, C012, C016, and C021. For each patient, slice-level predictions were generated by each model variant and then aggregated to obtain patient-level scores. Patient-level aggregation was used because axial slices from the same CT volume are correlated and should not be treated as independent observations in statistical testing. This design provides a more conservative evaluation than slice-level testing and reduces the risk of inflated statistical significance.

All models were evaluated using the same preprocessing and normalization workflow implemented in the local benchmark environment. The same image loading procedure, intensity normalization, windowing convention where applicable, and model inference pipeline were applied to all variants. This ensured that performance differences were attributable to architectural and objective-function differences rather than to inconsistent data handling.

Let x denote an LDCT input image and y denote the corresponding

normal-dose CT reference. Each model learns a mapping

$$\hat{y} = f_{\theta}(x), \quad (1)$$

where f_{θ} is the denoising network and $\hat{y}$ is the predicted denoised image. The goal is not only to minimize pixel-wise reconstruction error, but also to preserve local gradient structures that correspond to anatomical boundaries.

3.3 RED-CNN baseline

The baseline model was a RED-CNN-style residual encoder–decoder. RED-CNN processes the LDCT input through a sequence of convolutional encoder layers, compresses the representation into a bottleneck feature space, and reconstructs the denoised image through decoder layers. Skip or residual connections help transfer low-level image information from earlier layers to later reconstruction stages, reducing the risk of losing local structures during encoding.

RED-CNN was selected as the main comparator because it remains a compact, reproducible, and strong baseline for image-domain LDCT denoising. In this study, RED-CNN was not treated as a weak reference model. Instead, it was used as the primary benchmark to test whether a targeted edge-aware extension could improve edge preservation without making broad claims of overall superiority on all metrics.

3.4 Proposed EDR-REDNet architecture

EDR-REDNet retains the general RED-CNN encoder–decoder organization but introduces explicit edge-aware processing at the bottleneck. The architecture consists of four main stages: RED-CNN-style encoding, fixed Sobel edge extraction, edge-guided bottleneck fusion with edge-dilated residual blocks, and decoder-based image reconstruction.

Given an LDCT input x , the encoder first extracts feature maps:

$$F_b = E_{\theta}(x), \quad (2)$$

where E_{θ} denotes the encoder and F_b denotes the bottleneck feature representation. In parallel, a non-trainable Sobel module extracts edge maps from the same LDCT input. These edge maps are resized and projected to the bottleneck feature dimension, then fused with F_b . The fused representation is processed by edge-dilated residual blocks and passed to the decoder to produce the final denoised image:

$$\hat{y} = D_{\theta}(R_{\theta}(F_b + P_{\theta}(S(x)))), \quad (3)$$

where $S(\cdot)$ denotes the fixed Sobel edge extraction module, P_θ denotes a 1×1 projection layer, R_θ denotes the edge-dilated residual bottleneck module, and D_θ denotes the decoder.

Fixed Sobel edge-guidance layer

The `FixedSobelLayer` is a deterministic, non-trainable edge extraction module. It applies four Sobel filters to the LDCT input: horizontal, vertical, 45-degree diagonal, and 135-degree diagonal filters. The output is a four-channel edge representation:

$$S(x) = \{S_{0^\circ}(x), S_{90^\circ}(x), S_{45^\circ}(x), S_{135^\circ}(x)\}. \quad (4)$$

Because the Sobel kernels are fixed and registered as buffers rather than learnable parameters, this module does not increase the trainable parameter count. This design was selected for two reasons. First, Sobel filters provide transparent and interpretable edge cues in grayscale CT images. Second, fixed edge extraction avoids introducing another learned sub-network that might overfit to noise edges in a small training setting.

The extracted edge maps are resized to match the spatial resolution of the bottleneck feature maps. A 1×1 convolution projects the four-channel Sobel representation into the same channel dimension as the bottleneck features:

$$F_e = P_\theta(S(x)). \quad (5)$$

The projected edge feature F_e is then added to the bottleneck representation:

$$F_{\text{fused}} = F_b + F_e. \quad (6)$$

Additive fusion was used instead of channel concatenation to keep the decoder compact and to avoid substantial parameter growth after the bottleneck.

Edge-dilated residual bottleneck blocks

The edge-dilated residual bottleneck module is designed to increase contextual perception while maintaining feature resolution. Each block applies dilated convolution followed by local refinement and residual addition. Dilation rates of 2 and 3 are used to enlarge the receptive field without downsampling the bottleneck feature maps.

For an input feature map F , an edge-dilated residual block can be written as

$$R(F) = F + \phi_\theta(F), \quad (7)$$

where $\phi_\theta(\cdot)$ denotes the dilated convolutional transformation. In practice, ϕ_θ includes dilated convolution, non-linear activation, local convolutional refinement, and feature normalization where implemented in the codebase. The

residual addition stabilizes training and helps preserve the original bottleneck information.

The motivation for inserting these blocks at the bottleneck is that edge preservation requires both local gradient sensitivity and sufficient anatomical context. For example, a vessel boundary or fissure-like structure should be distinguished from stochastic noise. Dilated convolution provides a larger neighborhood for this decision without increasing the number of pooling operations or reducing spatial resolution.

Decoder and residual image reconstruction

After edge-guided fusion and edge-dilated bottleneck processing, the decoder reconstructs the denoised CT image:

$$\hat{y} = D_{\theta}(F_{\text{EDR}}), \quad (8)$$

where F_{EDR} is the output of the edge-dilated residual bottleneck. The decoder follows the RED-CNN-style reconstruction pathway, using learned convolutional operations to map latent features back to image space.

The output is trained to approximate the NDCT reference while preserving boundary structures. The architecture therefore remains an image-domain post-processing model: it does not require raw projection data, scanner reconstruction integration, or a differentiable sinogram-domain reconstruction layer. This makes it consistent with the practical motivation of RED-CNN-based LDCT denoising while adding explicit edge-structure guidance.

3.5 Objective function

The full EDR-REDNet model was trained using a combined reconstruction and edge-preservation loss:

$$L_{\text{total}} = L_{\text{Charbonnier}}(\hat{y}, y) + \alpha L_{\text{Sobel}}(\hat{y}, y), \quad (9)$$

where $\hat{y}$ is the predicted denoised image, y is the NDCT reference, and α controls the relative weight of the Sobel-gradient loss.

The Charbonnier loss is a smooth and robust reconstruction loss:

$$L_{\text{Charbonnier}}(\hat{y}, y) = \frac{1}{N} \sum_{i=1}^N \sqrt{(\hat{y}_i - y_i)^2 + \epsilon^2}, \quad (10)$$

where N is the number of pixels and ϵ is a small positive constant. Compared with MSE, the Charbonnier loss behaves similarly to L_1 loss while remaining differentiable near zero. It is less sensitive to large local outliers than squared

error and is therefore suitable for medical image restoration, where small high-contrast structures should not be dominated by a few large pixel errors.

The Sobel edge loss directly compares the gradient structures of the prediction and reference:

$$L_{\text{Sobel}}(\hat{y}, y) = \frac{1}{K} \sum_{k=1}^K \|S_k(\hat{y}) - S_k(y)\|_1, \quad (11)$$

where S_k denotes the k -th Sobel filter and $K = 4$ for the four edge directions. This loss encourages the model to preserve local gradient patterns and anatomical boundaries rather than optimizing only global pixel-wise similarity.

The combined loss was selected to align the optimization objective with the primary endpoint of the study. Since the intended contribution of EDR-REDNet is edge-structure preservation, direct gradient supervision is more appropriate than relying only on pixel-wise losses. This design also avoids the use of natural-image perceptual losses, such as VGG-based losses, which may not align with grayscale CT intensity physics or Hounsfield-unit behavior.

3.6 Ablation variants

A controlled ablation study was performed to isolate the contribution of each proposed component. Four variants were evaluated:

This design separates architectural edge processing from edge-supervised optimization. Variant B tests whether the edge-dilated bottleneck alone is sufficient. Variant C tests whether injecting Sobel edge information into the network representation improves edge preservation without modifying the loss. Variant D tests the complete design, combining edge-dilated contextual modeling, fixed Sobel input guidance, and Sobel-gradient supervision.

The ablation logic is important because an improvement in the full model should not be attributed to a single component without evidence. If Variant D outperforms Variants B and C on the edge-preservation endpoint, this suggests that both representation-level edge guidance and objective-level gradient supervision are required to obtain the observed improvement.

3.7 Evaluation metrics

Three image-quality metrics were used for quantitative evaluation: PSNR, SSIM, and Edge_SSIM.

Table 1. Comparison of EDR-REDNet architectural variants and their components.

Variant Description		Edge-dilated residual block	Sobel input guidance	Sobel-gradient loss	Purpose
A	RED-CNN baseline	No	No	No	Reference pretrained RED-CNN
B	RED-CNN + EdgeBlock	Yes	No	No	Tests dilated residual bottleneck alone
C	EdgeBlock + Sobel input guidance	Yes	Yes	No	Tests edge-map guidance without edge loss
D	Full EDR-REDNet	Yes	Yes	Yes	Tests combined architecture and edge-supervised loss

PSNR was used to quantify pixel-wise fidelity:

$$\text{PSNR} = 20 \log_{10} \left(\frac{I_{\max}}{\sqrt{\text{MSE}}} \right), \quad (12)$$

where $I_{\max}$ is the maximum intensity value after normalization and MSE is the mean squared error between the prediction and the reference. PSNR is widely reported in LDCT denoising, but it can favor overly smooth images because it strongly penalizes pixel-wise deviations.

SSIM was used to quantify global structural similarity between the denoised output and the NDCT reference. Although SSIM is more perceptual than PSNR, it still summarizes broad image similarity and may not sufficiently isolate local boundary preservation.

Edge_SSIM was used as the primary endpoint. Edge_SSIM applies structural similarity analysis to edge representations rather than directly to intensity images:

$$\text{Edge_SSIM} = \text{SSIM}(E(\hat{y}), E(y)), \quad (13)$$

where $E(\cdot)$ denotes an edge representation derived from Sobel or gradient-based filtering. This metric is better aligned with the purpose of EDR-REDNet because it directly evaluates whether the predicted image preserves edge structures similar to the reference. However, Edge_SSIM remains an image-quality surrogate and should not be interpreted as a clinical endpoint.

For qualitative edge analysis, gradient mean absolute error was also used in the zoomed regions:

$$\text{GMAE} = \frac{1}{N} \sum_{i=1}^N |E(\hat{y})_i - E(y)_i|, \quad (14)$$

where lower GMAE indicates closer agreement between the predicted and reference gradient structures. This measure was used to support visual interpretation in local lung-parenchymal regions and edge-error maps.

3.8 Patient-level statistical analysis

For each patient, slice-level PSNR, SSIM, and Edge_SSIM values were aggregated to obtain patient-level scores. The primary statistical comparison was then performed at the patient level, not at the slice level. This is essential because treating all slices as independent would overestimate the effective sample size and could produce misleading p-values.

The full EDR-REDNet model, Variant D, was compared with each comparator using a one-sided Wilcoxon signed-rank test. The test was selected because the sample size was small and normality could not be assumed.

The alternative hypothesis was

$$H_1 : \text{Edge_SSIM}_D > \text{Edge_SSIM}_{\text{comparator}}, \quad (15)$$

A p-value below 0.05 was considered statistically significant. The same testing framework was also applied to PSNR and SSIM as secondary analyses, but these metrics were not the primary basis for evaluating the proposed contribution.

This statistical design reflects the main scientific question of the study: whether the full edge-aware model consistently improves patient-level edge preservation over RED-CNN and partial variants.

3.9 Computational efficiency analysis

Model efficiency was evaluated using parameter count, MACs, and inference time for a 512×512 input image. Parameter count measures the size of the trainable model. MACs estimate the computational burden of a forward pass. Inference time measures practical runtime in the local hardware environment.

These efficiency metrics are important because LDCT denoising may be used in large-volume CT workflows, including lung screening, where a single patient scan can contain many axial slices. A structure-preserving method should therefore not require disproportionate computational resources relative to its benefit. In this study, the edge-aware modules were designed to introduce only moderate overhead compared with RED-CNN.

3.10 Qualitative edge-preservation protocol

In addition to global and patient-level metrics, representative lung-parenchymal regions were selected for qualitative visualization. The goal of this visualization was to show whether the Edge.SSIM improvement corresponded to visible preservation of vessel-like structures, fissure-like boundaries, and local parenchymal edges.

For each selected case, the same region of interest was displayed across the LDCT input, RED-CNN output, EDR-REDNet output, and NDCT reference. Edge-error maps were generated by comparing the gradient representation of each denoised output with the NDCT reference. Lower residual intensity in the edge-error map indicates closer local gradient agreement with the reference.

This qualitative analysis was intended to support the technical interpretation of the quantitative results. It was not designed as a radiologist

reader study and should not be interpreted as evidence of clinical diagnostic superiority. Instead, it provides visual evidence that the proposed model’s improvement in Edge_SSIM corresponds to local edge behavior in lung CT regions.

3.11 Reproducibility checklist

The experiment was implemented within the local ldct-benchmark project. The primary model variants were the RED-CNN baseline, EdgeBlock variant, Sobel-input partial variant, and full EDR-REDNet. The primary endpoint was patient-level Edge_SSIM. The primary statistical test was the one-sided Wilcoxon signed-rank test comparing Variant D against each comparator. The main result files included summary metric tables, per-patient scores, Wilcoxon p-values, model-efficiency results, and qualitative edge-preservation figures.

This reporting strategy follows the benchmark-oriented logic that a new LDCT denoising method should report not only average image-quality metrics, but also ablation behavior, patient-level consistency, statistical testing, computational cost, and qualitative evidence.

4 Results

4.1 Quantitative performance on global and edge-preservation metrics

Table 2 summarizes the quantitative performance of the four evaluated variants on the held-out test set. RED-CNN achieved the highest PSNR and global SSIM, whereas the full EDR-REDNet model achieved the highest Edge_SSIM. Specifically, RED-CNN obtained a PSNR of 44.0599 ± 6.7933 and SSIM of 0.9361 ± 0.0533 , while full EDR-REDNet obtained a slightly lower PSNR of 43.8469 ± 6.7838 and SSIM of 0.9326 ± 0.0561 . In contrast, Edge_SSIM increased from 0.7608 ± 0.1308 for RED-CNN to 0.7879 ± 0.1291 for full EDR-REDNet.

This result indicates a clear trade-off between conventional global fidelity metrics and the proposed edge-preservation endpoint. The RED-CNN baseline remains stronger under PSNR and global SSIM, which are mainly sensitive to pixel-wise and broad structural similarity. However, the full EDR-REDNet variant is superior under Edge_SSIM, suggesting that the proposed edge-guided architecture and Sobel-gradient supervision improve the preservation of local boundary structures. Therefore, the quantitative results should not be interpreted as evidence that EDR-REDNet provides better overall image quality than RED-CNN. Instead, they support the narrower conclusion that EDR-REDNet shifts the denoising behavior toward better edge-structure preservation.

Table 2. Summary performance on the held-out test set.

Variant	PSNR $\uparrow$	SSIM $\uparrow$	Edge_SSIM $\uparrow$
A — RED-CNN baseline	44.0599 ± 6.7933	0.9361 ± 0.0533	0.7608 ± 0.1308
B — + EdgeBlock	43.9909 ± 6.8479	0.9329 ± 0.0561	0.7598 ± 0.1265
C — + Sobel Input	43.8845 ± 6.8087	0.9333 ± 0.0556	0.7583 ± 0.1310
D — Full EDR-REDNet	43.8469 ± 6.7838	0.9326 ± 0.0561	0.7879 ± 0.1291

The most important observation is that the full model improves the metric most closely aligned with the intended contribution of this study. Since LDCT denoising may suffer from over-smoothing when optimized mainly for pixel-wise similarity, Edge_SSIM provides a more targeted evaluation of

whether vessel boundaries, fissure-like structures, and local gradient patterns are retained after denoising.

4.2 Patient-level Edge_SSSIM analysis

Because CT slices from the same patient are not statistically independent, patient-level aggregation was used for the primary endpoint. The full EDR-REDNet model improved Edge_SSSIM over RED-CNN in all nine held-out patients: L150, L110, L232, L134, C002, C004, C012, C016, and C021. This consistent direction of improvement is important because it shows that the gain is not driven by a single favorable case or isolated slice-level outliers.

Table 3. Patient-level Edge_SSSIM values.

Patient	RED-CNN	+ EdgeBlock	+ Sobel Input	Full EDR-REDNet
L150	0.924456	0.915496	0.920704	0.941806
L110	0.904371	0.899315	0.902707	0.930134
L232	0.893838	0.888865	0.890569	0.923711
L134	0.898956	0.895255	0.898346	0.927522
C002	0.628549	0.631889	0.627266	0.661672
C004	0.654224	0.653163	0.646606	0.678672
C012	0.635132	0.637832	0.631855	0.666545
C016	0.688048	0.690477	0.686768	0.717000
C021	0.619752	0.626168	0.619491	0.644286

The improvement pattern was observed in both higher-Edge_SSSIM cases and lower-Edge_SSSIM cases. For example, among the higher-scoring cases, L150 improved from 0.924456 to 0.941806, and L232 improved from 0.893838 to 0.923711. Among the more challenging chest cases, C004 improved from 0.654224 to 0.678672, C016 from 0.688048 to 0.717000, and C021 from 0.619752 to 0.644286. This supports the interpretation that the full model improves local edge fidelity across heterogeneous patient-level conditions rather than only in visually easy examples.

4.3 Statistical significance

One-sided Wilcoxon signed-rank testing confirmed that full EDR-REDNet significantly improved patient-level Edge_SSSIM compared with RED-CNN

and with both partial ablation variants. The Edge_SSIM comparison was significant for D versus A, D versus B, and D versus C, with $p = 0.0020$ in all three comparisons. In contrast, PSNR and global SSIM did not show statistically significant improvement for the full model.

Table 4. *One-sided Wilcoxon signed-rank tests.*

Comparison	PSNR p	PSNR result	SSIM p	SSIM result	Edge_SSIM p	Edge_SSIM result
D vs A	1.0000	Not significant	1.0000	Not significant	0.0020	Significant
D vs B	1.0000	Not significant	0.8750	Not significant	0.0020	Significant
D vs C	0.9980	Not significant	1.0000	Not significant	0.0020	Significant

These findings reinforce the main message of the study. EDR-REDNet should not be framed as a model that universally improves all image-quality metrics. Rather, it provides a statistically consistent gain on the predefined edge-preservation endpoint. This is important for LDCT restoration because an algorithm can achieve high PSNR or SSIM while still smoothing small boundaries and fine anatomical details. The present results suggest that adding explicit Sobel-based guidance and gradient supervision can redirect the network toward local structure preservation.

4.4 Computational efficiency

The proposed edge-aware components introduced a moderate computational overhead. Compared with RED-CNN, full EDR-REDNet increased the number of trainable parameters from 1.85M to 2.18M and increased MACs from 462.09G to 542.50G for a 512×512 input. Inference time increased from 200.71 ms for RED-CNN to 245.92 ms for full EDR-REDNet, corresponding to approximately $1.18\times$ parameters and $1.23\times$ runtime relative to the baseline.

This overhead is relatively modest given the additional edge-processing pathway and edge-dilated residual bottleneck. The result suggests that EDR-REDNet remains a compact extension of RED-CNN rather than a substantially larger transformer-scale or diffusion-based model. However, runtime should still be interpreted cautiously because practical deployment in lung screening would require evaluation on complete CT volumes, scanner-specific

Table 5. Model efficiency results for 512×512 input.

Model	Parameters	MACs	Time (ms)	Relative parameters	Relative time
A — RED-CNN	1.85M	462.09G	200.71	1.000	1.000
B — No Sobel	2.18M	542.40G	242.71	1.179	1.209
C — No EdgeBlock	1.85M	462.19G	204.15	1.000	1.017
D — EDR-REDNet	2.18M	542.50G	245.92	1.180	1.225

hardware, and batch-processing workflows.

4.5 Qualitative edge-preservation evidence

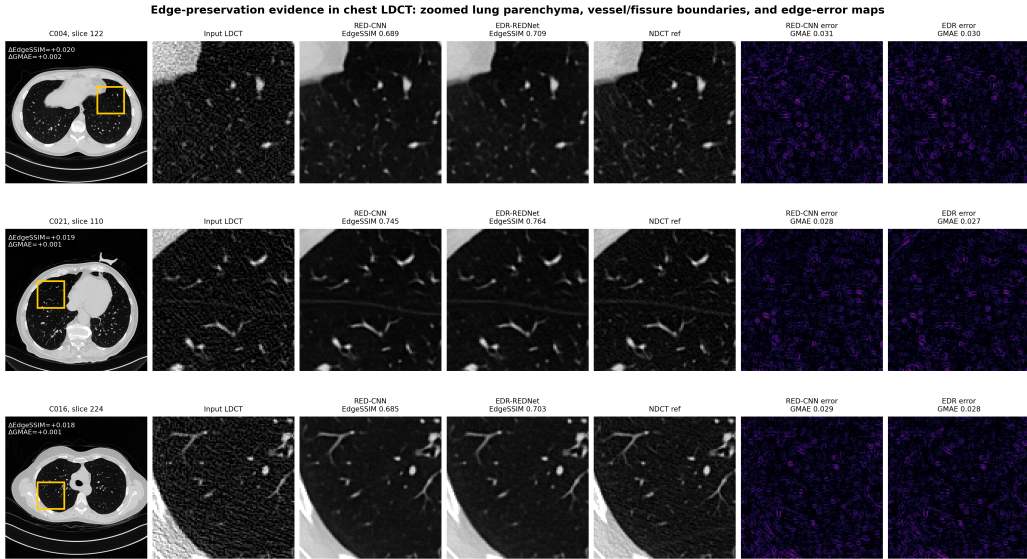


Figure 1. Edge-preservation evidence in representative chest LDCT cases.

Zoomed regions from three held-out chest CT cases are shown: C004 slice 122, C021 slice 110, and C016 slice 224. Each row displays the full CT slice with the selected lung ROI, LDCT input, RED-CNN output, EDR-REDNet output, NDCT reference, RED-CNN edge-error map, and EDR-REDNet edge-error map. EDR-REDNet increased local Edge_SSIM relative to RED-CNN in all three examples: +0.020 for C004, +0.019 for C021, and

+0.018 for C016. Gradient mean absolute error was also slightly reduced in the EDR-REDNet outputs. These zoomed examples support the quantitative patient-level finding that EDR-REDNet improves edge-structure preservation, particularly around lung parenchyma, vessel-like structures, and local boundary regions. The figure should be interpreted as qualitative technical evidence rather than a radiologist-validated clinical endpoint.

Figure 1 provides qualitative evidence for the structure-preserving behavior of EDR-REDNet in representative chest LDCT regions. The figure shows three held-out chest CT examples: C004 slice 122, C021 slice 110, and C016 slice 224. For each case, the panels include the LDCT input, RED-CNN output, EDR-REDNet output, NDCT reference, and corresponding edge-error maps. The selected regions focus on lung parenchyma, vessel-like structures, fissure or boundary regions, and local edge details.

In C004 slice 122, EDR-REDNet increased local Edge_SSIM from 0.689 to 0.709. The edge-error map also showed a small reduction in gradient mean absolute error, from 0.031 for RED-CNN to 0.030 for EDR-REDNet. In C021 slice 110, Edge_SSIM increased from 0.745 to 0.764, while the edge-error score decreased from 0.028 to 0.027. In C016 slice 224, Edge_SSIM increased from 0.685 to 0.703, and the edge-error score decreased from 0.029 to 0.028. These representative examples are consistent with the patient-level quantitative results and show that the improvement is visible in local lung structures rather than only in aggregate metrics.

Visually, the LDCT input contains strong noise that obscures small vascular and parenchymal details. RED-CNN suppresses noise effectively but produces smoother local structures in several zoomed regions. EDR-REDNet retains more boundary-like information around vessel and fissure structures while maintaining denoised appearance close to the NDCT reference. The edge-error maps make this difference more explicit: lower residual intensity in EDR-REDNet indicates closer agreement between predicted and reference gradient structures.

Importantly, these qualitative findings should be interpreted as edge-preservation evidence, not as proof of clinical superiority. The selected zooms demonstrate that EDR-REDNet can better preserve local gradient patterns in representative cases, but radiologist observer studies are still required to determine whether this improvement translates into better lesion visibility, diagnostic confidence, or nodule assessment.

4.6 Summary of findings

Overall, the experimental results support the central hypothesis of this study. Full EDR-REDNet improves the predefined edge-preservation endpoint, Edge_SSIM, over RED-CNN and partial ablation variants. The gain is consistent across all nine held-out patients and statistically significant under patient-level Wilcoxon testing. However, PSNR and global SSIM are slightly lower than RED-CNN and do not improve significantly. Therefore, the appropriate conclusion is that EDR-REDNet provides a structure-preserving trade-off: it sacrifices a small amount of global pixel-wise fidelity in exchange for more consistent edge preservation.

This interpretation is aligned with the intended design of the model. The FixedSobelLayer provides explicit edge guidance, the edge-dilated residual bottleneck increases local contextual modeling, and the Sobel-gradient loss directly supervises boundary structure. Together, these components make EDR-REDNet more sensitive to edge fidelity than RED-CNN while keeping model size and runtime within a moderate overhead range.

5 Discussion

5.1 Principal findings

This study proposed EDR-REDNet, an edge-aware extension of RED-CNN for image-domain LDCT denoising. The main finding is that the full EDR-REDNet model improved the predefined edge-preservation endpoint, Edge_SSIM, compared with the RED-CNN baseline and partial ablation variants. The improvement was consistent across all nine held-out patients and statistically significant under patient-level one-sided Wilcoxon signed-rank testing. In contrast, PSNR and global SSIM were slightly lower than those of RED-CNN and did not show significant improvement. Therefore, the results should be interpreted as evidence of a structure-preserving trade-off, rather than as evidence that EDR-REDNet provides universally superior denoising quality.

This interpretation is important for LDCT restoration. A denoising model optimized primarily for pixel-wise fidelity may suppress noise effectively but also smooth subtle high-frequency anatomical information. In chest CT, such high-frequency structures may include vessel margins, fissure-like boundaries, small airway contours, pleural interfaces, and lesion edges. The proposed model intentionally shifts part of the optimization objective from global intensity fidelity toward local gradient preservation. The resulting performance profile is therefore expected: RED-CNN remains stronger on PSNR and global SSIM, whereas EDR-REDNet is stronger on Edge_SSIM.

5.2 Structure-preserving trade-off

The central trade-off observed in this study is that EDR-REDNet improves edge-structure preservation while sacrificing a small amount of global pixel-wise fidelity. RED-CNN achieved the highest PSNR and global SSIM, whereas full EDR-REDNet achieved the highest Edge_SSIM. This finding is consistent with the known behavior of conventional full-reference metrics: PSNR is closely related to mean squared error and can favor smoother outputs, while global SSIM summarizes broad structural similarity and may not isolate small local boundary behavior.

This trade-off should not be treated as a weakness of the proposed model, but as the intended effect of the design. EDR-REDNet was not designed to maximize all conventional metrics. It was designed to test whether adding explicit edge input and Sobel-gradient supervision to a RED-CNN-style backbone can reduce over-smoothing and preserve boundary structures

more effectively. The results support this narrower hypothesis. They do not support a broad claim that EDR-REDNet is globally superior to RED-CNN under all image-quality criteria.

Recent LDCT denoising reviews emphasize that denoising should preserve image edges, maintain global contrast, avoid texture loss, and avoid introducing new artifacts. These requirements are broader than noise suppression alone and are particularly relevant in medical imaging, where subtle differences in highly similar anatomical backgrounds may indicate disease. In this context, Edge_SSIM provides a useful technical surrogate for the specific goal of edge preservation, even though it cannot replace reader studies or downstream clinical evaluation.

5.3 Role of the edge-aware components

The ablation results indicate that neither edge-dilated residual processing alone nor Sobel input guidance alone was sufficient to improve Edge_SSIM. Variant B, which introduced the edge-dilated residual bottleneck without Sobel guidance or Sobel-gradient loss, did not improve the primary endpoint. Variant C, which introduced Sobel input guidance without the complete edge-supervised objective, also did not improve Edge_SSIM. Only the full EDR-REDNet model, combining edge-dilated residual blocks, fixed Sobel guidance, and Sobel-gradient supervision, achieved consistent improvement.

This suggests that edge preservation requires both representation-level guidance and objective-level supervision. The fixed Sobel layer provides deterministic edge cues at the input-to-bottleneck pathway, whereas the Sobel-gradient loss forces the predicted image to align with the reference gradient structure. The edge-dilated residual blocks then provide additional contextual capacity at the bottleneck, allowing the network to interpret edge cues using a wider local neighborhood. This combination appears to reduce the tendency toward over-smoothing more effectively than any single component alone.

The design is related to earlier edge-aware LDCT denoising work. Gholizadeh-Ansari et al. used dilated convolutions, residual learning, a non-trainable edge detection layer, and perceptual loss to preserve structural detail and reduce over-smoothing. EDR-REDNet follows the same general motivation but differs in its RED-CNN-based benchmark positioning and in its use of Sobel-gradient supervision rather than natural-image perceptual features. This is a deliberate choice because CT images are grayscale medical images with quantitative intensity meaning, and Sobel gradients are directly interpretable as local boundary changes.

5.4 Interpretation of qualitative edge evidence

The qualitative zoom analysis supports the quantitative Edge_SSIM findings. In the representative chest CT cases, EDR-REDNet improved local Edge_SSIM relative to RED-CNN in all displayed examples: C004 slice 122, C021 slice 110, and C016 slice 224. The edge-error maps also showed slightly lower gradient error for EDR-REDNet than for RED-CNN. These examples focus on lung parenchyma, vessel-like structures, fissure or boundary regions, and local edge patterns.

The visual results suggest that RED-CNN suppresses noise effectively but can produce smoother local structures, whereas EDR-REDNet better maintains boundary-like patterns around selected pulmonary structures. However, these zoomed examples should be interpreted as technical evidence, not as clinical proof. The selected regions demonstrate that the proposed model can better preserve local gradient structure in representative cases, but they do not establish that radiologists would detect more nodules, assign higher diagnostic confidence, or make different clinical decisions.

Therefore, the qualitative evidence is best used to support the mechanism of the model: the Edge_SSIM improvement corresponds to visible local edge behavior. It should not be used to claim diagnostic superiority without reader studies.

5.5 Relation to benchmarking and reproducibility

The findings align with recent concerns in LDCT denoising benchmarking. Eulig et al. highlighted several weaknesses in the field, including inconsistent datasets, missing open-source implementations, unfair hyperparameter choices, and overreliance on PSNR, SSIM, and RMSE. They also noted that conventional metrics are often not well aligned with human reader assessment in medical imaging and recommended more rigorous evaluation, including clinically meaningful metrics, reader studies, and radiomic feature analysis.

This study responds to those concerns in a limited but focused way. It uses a controlled ablation design, patient-level aggregation, Wilcoxon testing, and efficiency reporting. It also avoids claiming broad superiority based only on mean slice-level metrics. The primary endpoint, Edge_SSIM, is aligned with the proposed model objective rather than selected post hoc. However, the present study remains narrower than a full benchmark evaluation. EDR-REDNet has not yet been compared with the complete ldct-benchmark model set under the same Edge_SSIM, radiomic, and reader-study endpoints.

A future submission would be stronger if EDR-REDNet were evalu-

ated against additional benchmark methods such as CNN10, WGAN-VGG, DU-GAN, TransCT, QAE, ResNet, and bilateral filtering under identical pre-processing and testing conditions. The benchmark literature also suggests that repeated training with multiple random seeds and standardized hyper-parameter optimization would improve the reliability of reported differences.

5.6 Relation to benchmarking and reproducibility

EDR-REDNet introduced a moderate increase in computational cost. The full model increased parameter count from 1.85M to 2.18M and inference time from 200.71 ms to 245.92 ms for a 512×512 input, corresponding to approximately $1.18\times$ parameters and $1.23\times$ runtime relative to RED-CNN.

This overhead is acceptable for research workflows and may be acceptable for offline denoising pipelines. The architecture remains compact compared with larger transformer-scale or diffusion-based models. However, runtime should be further assessed on full CT volumes, clinical hardware, and realistic batch-processing pipelines. Lung cancer screening and high-volume CT workflows may require processing hundreds of slices per examination, so slice-level runtime alone is insufficient for deployment evaluation.

The fixed Sobel guidance is computationally lightweight and does not substantially increase trainable parameters. Most of the added cost comes from the edge-dilated residual bottleneck. This suggests that future work could explore more efficient bottleneck designs, channel pruning, mixed-precision inference, or 2.5D input strategies that improve structure preservation without substantially increasing per-volume runtime.

5.7 Clinical and research implications

If validated further, edge-aware LDCT denoising may be useful in workflows where boundary visibility is important, including lung nodule assessment, vessel delineation, fissure visualization, organ boundary evaluation, and lesion margin analysis. However, Edge_SSIM improvement alone does not prove diagnostic superiority. It should be treated as a technical surrogate that motivates further clinical validation.

This distinction is especially important in lung cancer screening. LDCT screening has demonstrated mortality benefit in high-risk populations, but successful implementation depends on standardized imaging quality, nodule management, radiologist workflow, and AI support. A denoising model used in this setting must therefore be evaluated not only for visual quality, but also for lesion visibility, false-positive behavior, false-negative risk, quantitative HU stability, and downstream AI robustness.

From a research perspective, EDR-REDNet is useful because it provides a compact, interpretable, edge-aware modification to a RED-CNN-style baseline. Its fixed Sobel module and Sobel-gradient loss are simple to reproduce, and its computational overhead is moderate. This makes it suitable as a component for future benchmark studies, radiomic stability studies, and task-based evaluations

5.8 Limitations

This study has several limitations.

First, the test set contained only nine held-out patients. Although EDR-REDNet improved Edge.SSIM in all nine patients, this sample size is small and insufficient for broad generalization. Larger test cohorts are required to determine whether the improvement holds across scanners, dose levels, reconstruction kernels, anatomical regions, and patient body habitus.

Second, the primary endpoint was Edge.SSIM, which is an image-quality surrogate. It is better aligned with edge preservation than PSNR or global SSIM, but it does not directly measure diagnostic accuracy. No radiologist observer study was performed. Therefore, the present results cannot establish that EDR-REDNet improves lesion detection, diagnostic confidence, or clinical decision-making.

Third, the comparison set was limited to RED-CNN and controlled ablation variants. The study has not yet compared EDR-REDNet with a complete set of LDCT denoising algorithms under a standardized benchmark protocol. This limits the strength of claims relative to the broader literature.

Fourth, the evaluation was image-domain only. EDR-REDNet does not use raw projection data, sinogram-domain structure, or dual-domain physical consistency. This makes the method easier to deploy in settings where only reconstructed images are available, but it may also limit performance compared with methods that can exploit projection-domain information.

Fifth, downstream task performance was not evaluated. The effect of EDR-REDNet on lung nodule detection, segmentation, malignancy classification, radiomic feature stability, or cancer risk prediction remains unknown. Since denoising can alter texture and edge information, downstream validation is essential before clinical deployment.

Sixth, edge-aware losses may preserve noise edges as well as anatomical edges. The present results suggest improved agreement with reference gradient structure, but additional artifact analysis is required to ensure that the model does not amplify stochastic noise patterns, streak artifacts, or artificial sharp boundaries.

Finally, the current evaluation did not include multiple random seeds for all variants, extensive hyperparameter optimization, or external validation datasets. These steps are necessary to quantify training stability and reproducibility.

5.9 Future validation requirements

Several validation steps are required before EDR-REDNet can be considered clinically useful.

First, the model should be evaluated on larger multi-center datasets, including different scanner vendors, reconstruction kernels, dose levels, and anatomical regions. External validation is necessary to determine whether the edge-preservation gain generalizes beyond the current held-out cohort.

Second, EDR-REDNet should be compared with a broader set of LDCT denoising algorithms under the same preprocessing, split strategy, hyperparameter protocol, and Edge_SSIM/radiomic endpoints. This comparison should include both classical and deep learning methods, as recommended by recent benchmark studies.

Third, radiologist reader studies should be conducted. These should assess subjective noise, sharpness, lesion visibility, diagnostic confidence, and possible artifact perception. For chest LDCT, reader studies should include nodule-like structures, small vessels, fissures, ground-glass opacities, and pleural interfaces.

Fourth, task-based validation should be performed. This includes evaluating nodule detection sensitivity, false-positive rate, segmentation consistency, radiomic feature reproducibility, and downstream cancer risk prediction before and after denoising.

Fifth, future work should investigate adaptive edge losses that distinguish anatomical boundaries from stochastic noise edges. This may reduce the risk of preserving or enhancing non-anatomical gradients.

Sixth, 3D or slice-context extensions should be explored. The current model operates in a 2D image-domain setting. However, chest CT is volumetric, and inter-slice consistency is important for radiologist interpretation and downstream AI. A 2.5D or lightweight 3D extension may improve structural continuity while maintaining manageable computational cost

6 Conclusion

6.1 Principal findings

EDR-REDNet is a compact edge-aware extension of RED-CNN for image-domain LDCT denoising. By combining fixed four-direction Sobel edge guidance, edge-dilated residual bottleneck processing, and Sobel-gradient supervision, the model improves the predefined edge-preservation endpoint, Edge-SSIM, compared with RED-CNN and partial ablation variants. The improvement was observed in all nine held-out patients and was statistically significant under patient-level Wilcoxon testing.

The study does not show that EDR-REDNet is globally superior to RED-CNN on all image-quality metrics. RED-CNN achieved slightly higher PSNR and global SSIM. Instead, the findings demonstrate a meaningful structure-preserving trade-off: EDR-REDNet sacrifices a small amount of global pixel-wise fidelity while improving local edge-structure preservation.

These results support edge-aware architectural guidance and Sobel-gradient supervision as promising strategies to reduce over-smoothing in LDCT denoising. However, EDR-REDNet should be viewed as a technical contribution requiring further validation, not as a clinically validated denoising solution. Larger multi-center evaluation, comparison with full benchmark model sets, radiologist observer studies, radiomic stability analysis, and downstream task validation are required before clinical claims can be made.

References